Departement Elektriese, Elektroniese en Rekenaar-Ingenieurswese

Department of Electrical, Electronic and Computer Engineering

Semestertoets 3

Kopiereg voorbehou

Semester Test 3

Module EBN122

Elektrisiteit en Elektronika

November 2016

Module EBN122

Electricity and Electronics

November 2016

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Exam information: **Eksamensinligting:**			
Maximum marks: Maksimum punte:	45	*Full marks:* Volpunte:	46
Duration of paper: Duur van vraestel:	*90 minutes* 90 minute	*Open / closed book:* Oopboek / toeboek:	*Closed* Toe
Total number of pages (including this page): Totale aantal bladsye (hierdie blad ingesluit):		13	

IMPORTANT **- BELANGRIK**

1. *The examination regulations of the University of Pretoria apply.*
   Die eksamenregulasies van die Universiteit van Pretoria geld.
2. *All questions must be answered on the ANSWER SHEET.*
   Alle vrae moet op die ANTWOORDBLAD beantwoord word.
3. *Students may do their own rough calculations on the question paper – the question paper will not be taken in.*
   Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
4. *Answers must include units!*
   Antwoorde moet eenhede insluit!
5. ***Final answers must be written as real numbers, and not as fractions.***
   Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.

Dosent(e): *Lecturer(s):*	1. D le Roux 2. R dos Santos 3. J Joubert

EBN122 SEMESTER TEST/ TOETS 2 (NOV 2016)
ANSWER SHEET / ANTWOORDBLAD

Surname: Van:		*Student number:* Studentenommer:				
Initials: Voorletters:		*Group:* Groep:	ENGLISH (Eng 3-1)	ENGLISH (Eng 3-6)	AFR	

Q1+2	14	Q3	16	Q4	16	**Tot:**	46 /45

VRAAG 1 – QUESTION 1 (4)		
1	a) $V_1 =$ 2V ✓✓	b) $\alpha =$ 4 ✓✓

QUESTION 2 - VRAAG 2 (10)		
2.1	$i_{C(2\,s)} =$ 1.36mA ✓✓	
2.2	$i_{L(0\text{ to }6\,s)} =$ 4.8A ✓✓	
2.3	$C_{eq} =$ 3F ✓✓	
2.4	$w_C =$ 0.36 J = 360mJ ✓✓	$w_L =$ 24 J ✓✓

VRAAG 3 – QUESTION 3 (16)		
3.1	a) $\bar{V} = -21.21 + j11.21$ V ✓✓ (rectangular)	
	b) $\bar{Z}_L =$ 50∠90° Ω ✓✓ (polar)	
	c) $i_c(t) = 5\cos(5000t + 110°)$ A ✓✓	
3.2	a) $\bar{I}_1 = 1 + j5$ A ✓✓ (rectangular)	b) $\bar{V}_2 =$ 0.79∠108.4° V ✓✓ (polar)
3.3	$\bar{Z}_{ab} = 20 - j60$ Ω ✓✓ (rectangular)	
3.4	a) $\bar{I}_a = 1.5 - j3$ A ✓✓ (rectangular)	b) $\bar{I}_a =$ 0.74∠92.9° A ✓✓ (polar)

VRAAG 4 – QUESTION 4 (16)		
4.1	$a_1 + ja_2 =$ 2 + j3 ✓	$b_1 + jb_2 =$ 0 + j(-3) ✓
	$c =$ 3 ✓	$d =$ 8 ✓
4.2	$e_1 + je_2 =$ 5 + j ✓	$f_1 + jf_2 =$ 4 + j(-2) ✓
	$g =$ 1 ✓	$h =$ −1 ✓
4.3	$\bar{V}_{TH} =$ 4.8 + j2.4 V ✓✓ (rectangular)	$Z_{TH} =$ (2.4 − j2.4) kΩ ✓✓ (rectangular)
4.4	$\bar{I}_N =$ 4A ✓✓ (rectangular)	$Z_N =$ 2.828∠45° Ω ✓✓ (polar)